REGULATORY HYSTERESIS, EXTENDED PHENOTYPE, AND THE LEX ARTIS OF INSTITUTIONAL DESIGN:

How Synthetic Agent Simulations Transform State Liability for Defective Normative Production

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Abstract

This paper develops a unified theory of regulatory hysteresis, state liability for defective normative production, and the legal standard of care applicable to legislative activity. Three theoretical traditions are integrated: quality management (Deming 1982; Shewhart 1931), which establishes that defects must be prevented in the production process rather than detected after the fact; hysteresis theory (Blanchard and Summers 1986; Palley 2017), which formalizes the irreversibility of institutional deformation; and Extended Phenotype Theory (EPT, Lerer 2025a), which explains how legal memes construct institutional machinery that persists after the originating norm is repealed. The paper introduces the Regulatory Hysteresis Index (RHI), proposes a graduated strict liability framework based on multilevel intentionality analysis (Dennett 1987), and develops a central argument that has not previously appeared in the literature: the publication of Shapira et al. (2026, arXiv:2602.20021) establishes a scientific inflection point after which the omission of pre-legislative evolutionary simulation constitutes negligence in the normative production process. Just as the diffusion of ecography, radiology, and pre-surgical laboratory testing generated a new standard of medical care, the demonstrated availability of synthetic agent laboratories for institutional dynamics testing generates a new standard of legislative care. Legislators and regulators who approve high-impact norms (RHI > 0.50) after February 2026 without documented evolutionary simulation analysis cannot invoke ignorance of available diagnostic tools. The paper derives seven falsifiable predictions, proposes a fourth RHI dimension incorporating destroyed option value, and develops an extension of the Francovich doctrine applicable to domestic hysteretic damage.

Keywords: regulatory hysteresis; extended phenotype theory; state liability; lex artis; synthetic laboratory; evolutionary game theory; quality management; constitutional lock-in; option value; multilevel intentionality

JEL Codes: K00, K40, H11, D02, L15, C73

I. Introduction: The Missing Quality Control in Normative Production

A bolt manufacturer operating in Argentina must comply with dimensional specifications measured in hundredths of a millimeter, document every stage of the production process, apply statistical process control to measure variation, analyze root causes of defects, and submit to external quality audits. If a defective batch reaches the market, it is recalled. The damage is quantifiable. The liability is clear.

The legislature that regulates that manufacturer produces legislation without measurable outcome specifications, without statistical process control, without systematic feedback from norm users, without audits of the normative product, and without recall mechanisms. A defective norm is not recalled: it generates institutional infrastructure that persists indefinitely, deforms the environment in which subsequent norms must operate, and accumulates irreversible damage that no reform can fully reverse.

This asymmetry, though obvious once stated, has received remarkably little systematic attention in legal scholarship. The OECD has worked on regulatory quality for decades, but its framework operates at the level of political-institutional governance: regulatory impact analysis, public consultation, administrative simplification. These are process checklists, not quality management systems in the sense that Deming (1982) and Shewhart (1931) gave to manufacturing. ISO has developed over 22,600 standards for virtually every imaginable product and service. None addresses the quality of the regulatory product itself.

The consequences extend far beyond inefficiency. Every amparo filing, every declaration of unconstitutionality, every judicial reinterpretation that contradicts legislative intent represents a defect detected after the normative product has been released into the institutional environment. But unlike a defective bolt, which can be recalled and replaced, a defective norm generates institutional infrastructure that persists indefinitely. Specialized courts interpret it. Scholars write about it. Law schools incorporate it into curricula. Lawyers specialize in litigating under it. Administrative agencies build procedures around it. This institutional infrastructure does not disappear when the norm is repealed. This paper calls that phenomenon regulatory hysteresis.

The argument proceeds in nine sections. Section II establishes the theoretical framework integrating quality management, hysteresis theory, and EPT. Section III analyzes the EPT mechanism of phenotypic construction and parasitic manipulation. Section IV introduces the evolutionary game theory (EGT) multilevel framework for the legislative production chain. Section V formalizes the Regulatory Hysteresis Index. Section VI develops the central new argument of this revision: how the publication of Shapira et al. (2026) on autonomous agent systems establishes a scientific inflection point that transforms the lex artis applicable to legislative activity. Section VII proposes the graduated strict liability framework. Section VIII presents seven falsifiable predictions. Section IX draws policy implications. Section X concludes.

II. Theoretical Framework: Three Converging Traditions

2.1 Quality Management and Normative Production

W. Edwards Deming articulated fourteen principles for quality management transformation, several of which translate with structural precision to normative production. His first principle, creating constancy of purpose toward improvement of product, implies that the legislator should define, before drafting, what measurable outcome the norm must produce and commit to verifying it. The vast majority of Argentine legislation lacks outcome indicators. Laws are enacted by declared intention, not by functional specification.

His third principle, ceasing dependence on inspection to achieve quality, has an exact legislative parallel: the Argentine legal system relies on ex post inspection (constitutional review, judicial review, amparo) rather than building quality into the legislative process. His fifth principle, improving constantly and forever the system of production, collides with the reality that sunset provisions fail systematically because norms lack Plan-Do-Check-Act cycles. They are planned, executed, but rarely are results verified against original specifications, and almost never is adjustment made based on evidence.

Shewhart's concept of statistical process control offers a complementary analytical lens. If the same norm produces radically different results depending on the tribunal applying it, that constitutes variation outside statistical control. This variation can be measured and charted to demonstrate that certain norms operate outside specification in terms of outcome predictability. The absence of any such measurement in legislative practice represents a systematic quality management failure with identifiable, proportional liability consequences.

2.2 Hysteresis: From Physics to Institutions

In materials science, hysteresis describes a precise phenomenon: when a force is applied to a material and then removed, the material does not return to its original state. A residual deformation persists. The forward path (applying force) differs from the return path (removing force). The system's state depends on its history, not solely on present conditions.

In economics, Blanchard and Summers (1986) established the term for European labor markets: temporary unemployment became structural because workers lost skills, insiders protected their positions, and the NAIRU recalibrated upward. Palley (2017) formalized three hysteresis mechanisms in economic policy: exit costs, destruction of social capital (Humpty-Dumpty hysteresis, where organizational capital once destroyed cannot be reconstructed), and institutional inertia. Di Tella and MacCulloch (2006) demonstrated institutional hysteresis comparing Europe with the United States across the 1970s oil shocks.

None of these authors, however, reaches the mechanism that connects hysteresis to normative production. The missing link is the extended phenotype.

Regulatory hysteresis is qualitatively distinct from two related concepts that require explicit separation. Path dependence describes systems where history matters for current outcomes but does not necessarily imply irreversibility: a path-dependent system may eventually converge to the same equilibrium from different starting points. Institutional lock-in, measured by the Constitutional Lock-in Index (Lerer 2025b), describes resistance to change: the system blocks reform attempts. Lock-in says the system resists change. Hysteresis says something worse: even when change is achieved, the system remains permanently deformed. It does not return to the prior state.

2.3 The Extended Phenotype Mechanism

In Dawkins's (1982) original formulation, the phenotype of an organism extends beyond its body to include environmental effects caused by the organism's genes: beaver dams, spider webs, caddisfly cases. The critical insight is that the gene is the unit of selection, and its phenotypic effects on the environment are as much products of natural selection as morphological features.

When a legislator enacts a norm, that norm generates extended phenotype from day one. Courts interpret it. Doctrinarians write about it. Law schools incorporate it into curricula. Lawyers specialize in litigating under it. Public officials build administrative procedures around it. Each component cooperates with the others in a decentralized system without central coordination: judges cite the doctrine that law schools produce; law schools teach the jurisprudence that judges produce; professional associations defend the specialized jurisdiction that generates work for their members. This is not conspiracy. It is the ordinary operation of cultural replication under selection pressure.

The critical insight for hysteresis is this: if, five years later, the originating norm is repealed, the extended phenotype persists. The courts continue to exist. The specialists continue to need work. The doctrine remains published. The administrative procedures continue by inertia. The institutional material is permanently deformed. Any new norm enacted in that space is born conditioned by the infrastructure of its predecessor. This is the mechanism connecting EPT to hysteresis: the extended phenotype is the residual deformation that persists after the force (the norm) is removed.

III. The EPT Mechanism: Parasitic Manipulation and Niche Construction

3.1 Parasitic Memes and the Legislative Host

Experimental research by Hafer (2019) confirms that host manipulation by parasites constitutes an inheritable and selectable extended phenotype: the parasite Schistocephalus solidus alters its copepod host's behavior to maximize its own transmission, and this manipulation responds to artificial selection in only three generations. The parasite does not improve the host's survival; it exploits the host's behavioral machinery to maximize its own replication.

The analogy to normative production is structural, not metaphorical. Legal memes that generate hysteresis function as parasitic replicators: they capture the legislative host's production machinery to ensure their own replication, independently of whether the resulting norm improves social welfare. The individual legislator does not act to maximize social welfare; she replicates the memes she has internalized through professional socialization, ideological networks, and interest group pressure. The result is norms optimized for memetic fitness, not functional quality.

This explains three otherwise puzzling patterns. First, regulatory hysteresis persists even when virtually all actors acknowledge its dysfunction: the extended phenotype of the parasitic meme (the coalition of specialists, courts, doctrines) maintains the institutional environment hostile to replacement memes. Second, reform attempts are resisted by what appears to be an institutional immune system: narratives of acquired rights, the progressivity principle misapplied, constitutional status of the originating norm. Third, norms that were initially protective become extractive over time, a pattern consistent with Hafer's observation that the parasite's manipulation strategy shifts as the host's life stage changes.

3.2 Niche Construction and Autocatalytic Accumulation

Odling-Smee et al. define niche construction as the process by which organisms modify their own selective environments and those of other organisms. Applied to regulation, each norm that accumulates modifies the institutional niche in which subsequent norms must survive. Regulatory complexity generates selection pressure that favors memes capable of navigating that complexity, specifically compliance specialists, interpretive specialists, and administrative procedure designers, over memes that reduce it, specifically deregulatory proposals, sunset clauses, and simplification initiatives.

This generates a self-reinforcing cycle: hysteresis favors more hysteresis. The niche constructed by accumulated regulation selects against the very memes that could reverse the accumulation. This is not a temporary equilibrium that will self-correct through rational institutional updating; it is a stable attractor in the dynamical systems sense. The work of Itao and Kaneko (2025) on evolutionary dynamical-systems game theory formalizes this: niche-constructing systems converge to limit-cycle attractors where institutional arrangements generate temporal regularities that function as self-sustaining institutions. Escaping such attractors requires exogenous perturbation, the institutional annealing discussed in Section IX.

3.3 Coalitions as Groups Under Cultural Selection

De Araújo (2021) demonstrates that shared use of extended phenotypes generates indirect genetic effects in individuals who did not construct them, and that group selection emerges naturally from this shared use. Legislative norms are extended phenotypes of shared use by definition: all subjects of a jurisdiction inhabit the normative extended phenotype constructed by the legislature. The damage from regulatory hysteresis is, in EPT terms, the cost of forced shared use of a defective extended phenotype.

Richerson et al. (2016) establish that cultural group selection plays an essential role in explaining large-scale human cooperation. The mechanisms that stabilize inter-group differences, conformist social learning, coordination payoffs, punishment of deviance, symbolic markers, apply directly to legislative coalitions. Parliamentary blocs are groups in the Cooney (2023) sense: they compete for agenda dominance (inter-group selection) while each member maximizes political survival within the bloc (intra-group selection). The extended phenotype of a legislative coalition is its accumulated normative production. Successful coalitions are imitated by others; their normative strategies spread across the institutional environment.

The implication for hysteresis is that the accumulation of dysfunctional regulation is not a failure of individual rationality. It is a predictable outcome of selection dynamics operating simultaneously at two levels: individual legislators optimizing political survival produce norms that generate institutional infrastructure; legislative coalitions competing for agenda dominance favor visible, constituency-building regulation over invisible, deregulatory simplification. The result is the systematic accumulation documented in empirical data: Australia's compliance costs reached AU$160 billion (5.8% of GDP) in 2024, up from AU$65 billion (4.2%) in 2013; Argentina's CLI of 0.87 correlates with zero sustained labor reforms across 23 attempts.

IV. The Legislative Production Chain as a Multilevel Selection System

4.1 Cooney's Framework Applied to Legislation

Cooney (2023) uses partial differential equations to model selection dynamics simultaneously within and between groups. The central tension is a tug-of-war between intra-group selection (individual incentive to defect, free-riding) and inter-group selection (collective incentive to cooperate when better-regulated jurisdictions outcompete worse-regulated ones). Crucially, the population structures most conducive to cooperation in multi-scale systems differ from those optimal within a single group.

Applied to the legislative production chain, each level of the system faces different selection pressures. Individual legislators optimize for reelection (intra-group). Parliamentary blocs optimize for agenda dominance (inter-group within the legislature). The executive branch optimizes for governability and political survival (cross-institutional). Technical advisors optimize for professional advancement within their institutional context (intra-level). None of these optimization targets aligns with minimizing regulatory hysteresis.

The seven mechanisms that promote cooperation under multilevel selection, assortment, reciprocity, population structure, punishment, coordination, selective migration, and imitation of successful groups, are all either absent from or distorted in the legislative production context. Selective migration toward better-regulated jurisdictions operates, but slowly (3-5 year lag in capital and talent flows). Punishment of legislative negligence is structurally blocked by sovereign immunity doctrines. Imitation of successful jurisdictions is mediated by political filters that favor ideological compatibility over regulatory quality. The result is a system that systematically under-produces quality and over-produces hysteresis.

4.2 Multilevel Intentionality in the Legislative Chain

Dennett's (1987) taxonomy of intentionality levels provides the micro-foundation for graduated liability. Different actors in the legislative production chain operate at radically different cognitive levels.

The individual legislator functions, in principle, as a Level 3 agent. She has beliefs about voters' beliefs, about other legislators' intentions, about how judges will interpret the norm. She can foresee consequences, anticipate objections, model reactions. If a legislator votes for a law knowing, or being expected to know, that it generates irreversible institutional hysteresis, she has the cognitive capacity to understand the damage. The question is whether she chose not to evaluate it.

The parliamentary bloc or legislative coalition operates as a hybrid system, between Level 1 and Level 2. The coalition optimizes votes, negotiates packages, trades support. It processes binary positions (approve/reject) on texts that few members read in full. Bloc behavior is more algorithmic than reflective. It does not have capacity to evaluate systemic effects like hysteresis, but it does have the institutional responsibility to incorporate evaluation mechanisms that do.

The executive branch, when submitting a bill, operates as a corporate system: Level 1 disguised as Level 3. It has declared mission (general welfare), Level 3 language in presidential messages invoking the common good, but its decisional machinery optimizes political variables without recursively modeling the mental states of those affected. This is the same gap identified in the Dennett-Nash framework applied to corporate criminal liability (Lerer 2025c): declared intentionality does not equal operative intentionality.

The technical legislative apparatus, advisors, committees, secretariats, should function as Level 3 injectors into a process that tends toward Level 1. They are the actors who could perform regulatory hysteresis impact assessment and evolutionary simulation. But in practice, they are subordinated to the bloc's political logic, so their Level 3 capacity is structurally neutralized.

This multilevel structure has direct implications for liability. The standard of diligence must scale according to each actor's position in the intentionality hierarchy and according to the tools available to them. A committee chair who directed consideration without ordering hysteresis assessment bears greater responsibility than a backbench legislator who followed party instructions. The executive that submitted the bill without evolutionary impact analysis bears design responsibility for the normative product. The technical apparatus that failed to flag available simulation tools bears professional responsibility for the quality failure.

4.3 The Olson Problem: Concentrated Benefits, Diffuse Costs

The Logic of Collective Action (Olson 1965) explains why regulatory hysteresis is politically stable despite its aggregate costs. The benefits of each specific norm are concentrated in small, organized groups: regulated industries, professional associations, specialized practitioners, administrative agencies whose budget grows with regulatory complexity. The costs of regulatory accumulation are diffuse across the entire population, generating a classic collective action problem.

The result is a coalition of extractors (regulated incumbents, compliance industry, regulatory bureaucracy) that actively benefits from the hysteretic equilibrium and resists perturbation. In EGT terms, this coalition plays a coordinationequilibrium strategy: each member benefits from the others' investment in maintaining the status quo, and defection (supporting reform) imposes costs without proportional private benefit. The coalition is evolutionarily stable precisely because its members have built extended phenotypes, specialized courts, professional organizations, academic positions, that generate ongoing selective pressure against the memes that would eliminate the coalition's institutional rents.

V. The Regulatory Hysteresis Index: Formalization and Extension

5.1 Base Formula

The Regulatory Hysteresis Index measures the irreversible institutional infrastructure generated by each norm. The base formula integrates five dimensions:

$$RHI = 0.30J + 0.25P + 0.20D + 0.15T + 0.10A$$

Where: J = jurisdictional creation (0-1), measured by specialized courts and tribunals established; P = professional specialization (0-1), measured by licensed practitioners whose livelihoods depend on the norm; D = doctrinal elaboration (0-1), measured by academic commentary, treatises, and university curricula; T = treaty integration (0-1), measured by international obligations incorporated; A = administrative embedding (0-1), measured by procedures, databases, and organizational units built around the norm.

The weights reflect the relative permanence of each institutional dimension. Jurisdictional creation receives the highest weight (0.30) because courts, once established, are the most difficult institutional structures to dissolve: they require legislative action, generate their own constituency among judges and lawyers, and their elimination is framed politically as an attack on judicial access. Professional specialization ranks second (0.25) because practitioners whose livelihoods depend on a norm constitute a politically mobilized constituency against repeal.

5.2 Extension: Destroyed Option Value

Arrow and Fisher (1974) developed the concept of option value: when a decision is irreversible and uncertainty about future consequences is substantial, preserving the capacity to act in the future has independent value beyond the expected value of current action. Sunstein (2002) applied this to regulatory law: before irreversible harms, regulators should 'buy an option' by investing in reversible precautionary measures rather than irreversible commitments.

Legislative hysteresis destroys option value systematically. Each norm that accumulates without review mechanism eliminates the capacity to adopt alternative regulatory approaches in the future. High-RHI norms are particularly damaging: they not only produce their immediate effects but preempt the entire space of regulatory alternatives that the institutional infrastructure makes inaccessible. This destroyed option value constitutes an independent dimension of damage.

The extended formula incorporates this dimension:

$$RHI_e = 0.28J + 0.23P + 0.18D + 0.14T + 0.09A + 0.08 \cdot OV$$

Where OV represents destroyed option value, estimated as the reduction in the feasible space of regulatory alternatives directly attributable to the norm's institutional infrastructure, scored 0-1. The weight reduction on other dimensions (approximately 0.02 each) reflects the reallocation to the new component while preserving the total to 1.00.

5.3 Empirical Calibration

Jurisdiction / Regime	J	P	D	T	RHI (base)
Argentina — Labor (Art. 14bis + LCT)	0.95	0.92	0.88	0.90	**0.91**
Brazil — Labor (pre-reform 2017)	0.70	0.62	0.58	0.55	**0.55**
Chile — Labor (post-reform)	0.30	0.28	0.25	0.22	**0.31**
Spain — Labor (pre-2012)	0.55	0.50	0.48	0.60	**0.44**

The correlation between RHI and reform success is striking: Argentina's 0.91 corresponds to zero sustained reforms across 23 attempts; Chile's 0.31 corresponds to six successful reforms in the same period; Brazil's pre-reform 0.55 corresponded to one reform achieved with significant implementation resistance; Spain's 0.44 corresponded to the 2012 reform that partially succeeded but generated the second-largest constitutional challenge in post-Franco history.

VI. The Lex Artis Revolution: How Agents of Chaos Changes the Standard of Care

6.1 The Lex Artis as a Dynamic Standard

The lex artis, the standard of care applicable to a profession, is not static. It incorporates available tools as they become demonstrably reliable and accessible. A physician in 1930 was not negligent for failing to perform an MRI; the technology did not exist. A physician in 1985 was not negligent for failing to order a CT scan for every headache; the technique was new, expensive, and not yet standard. A physician in 2026 who fails to request a CT scan when symptoms clearly indicate the need is negligent: the tool is available, reliable, and its omission falls below the standard that a diligent professional is expected to satisfy.

The same logic governs legislative design. Until recently, a legislator or executive facing a proposed norm could plausibly argue: 'No tools existed to simulate the evolutionary behavior of a regulation before implementation. We could not have known the institutional deformation our norm would generate.' That argument was always imperfect: agent-based modeling has existed since the 1990s; regulatory impact assessment has been an OECD standard since 2008; evolutionary game theory offered formal predictions about institutional dynamics. But it retained some force.

That argument collapsed on February 23, 2026.

6.2 The Shapira et al. Inflection Point

Shapira et al. (2026), Agents of Chaos (arXiv:2602.20021), published the results of a two-week red-teaming study of LLM-powered autonomous agents deployed in a realistic institutional environment with persistent memory, email, Discord, file systems, and shell execution. Twenty researchers interacted with five agents under benign and adversarial conditions, documenting eleven representative failure cases.

The authors' stated purpose was to identify security, privacy, and governance vulnerabilities in autonomous agent systems. That purpose is legitimate and urgent. But the paper simultaneously constitutes independent empirical evidence for a proposition with direct implications for legislative liability: that populations of synthetic agents with persistent memory and evolutionary time replicate, in laboratory conditions, the same institutional dynamics that EPT predicts for human normative systems.

The specific correspondence is precise. The paper documents agents trapped in resource consumption loops with no exit mechanism, which is the functional analog of regulatory lock-in. It documents instructions propagating from agent to agent through a process of shared-use institutional adoption, which is the functional analog of norm diffusion through the extended phenotype network. It documents "constitutional" instructions replaced by parasitic alternatives that respect the protocol syntax while replacing the objectives, which is the functional analog of doctrinal capture described in EPT. It documents escalating remedial actions that worsen the original problem, which is the functional analog of the administrative ratchet. And it documents the systematic absence of group-level optimization despite institutional machinery designed to produce it, which directly validates EPT's critique of multilevel selection theory applied to legal institutions.

None of these findings was incidental. They were reproducible, varied systematically across case types, and the authors explicitly identified them as structural rather than contingent failures of the systems studied. The paper establishes, in peer-reviewed preprint form from seventeen institutions including MIT, Harvard, Stanford, and Northeastern, that synthetic agent populations with evolutionary time exhibit institutional dynamics that follow the same laws as biological and cultural evolution.

6.3 The Elevator Inversion: A New Standard of Care for Legislative Design

In Argentine tort law, a traditional principle limits building liability for elevator accidents when an accessible staircase exists: the building bears responsibility for making a safe staircase available; it does not bear responsibility if a user chooses the elevator and the elevator fails, provided the elevator was reasonably maintained and the user was not compelled to use it.

The principle inverts when applied to diagnostic tools in professional practice. A physician who performs surgery without ordering pre-surgical tests is not relieved of liability by arguing that the surgery could have proceeded safely via the staircase route (general observation, clinical intuition). The availability of the diagnostic tool changes the standard: using the stairs when the elevator is accessible and the destination is high-risk is itself the negligence.

Regulatory simulation is the elevator. The staircase is traditional legislative process without diagnostic tools. The destination is any norm with RHI > 0.50.

Before Shapira et al. (2026), the simulation elevator was under construction: available in prototype, but not peer-reviewed, not institutionally validated, not documented in the scientific literature as reliable for institutional dynamics analysis. After February 23, 2026, the elevator is documented, demonstrated, and accessible. Regulators and legislators designing norms with substantial institutional impact who proceed without evolutionary simulation analysis are now using the stairs when the elevator is available and the destination is high-risk.

The standard of care for normative production has changed. It has changed not because a legislature passed a statute, not because a court issued a ruling, but because the scientific community published evidence that the diagnostic tool works.

6.4 The Medical Analogy: Three Parallel Omissions

The medical malpractice analogy is the most analytically precise because Argentine jurisprudence has developed detailed doctrine on the lex artis standard and its evolution with available technology. Three specific omissions correspond to identifiable legislative negligence categories.

Omission 1: Operating Without Pre-Surgical Tests (Norm Without Hysteresis Impact Assessment)

A surgeon who proceeds to operate without ordering pre-surgical tests that are standard for the type of intervention is negligent regardless of the operation's outcome. The negligence is in the omission of the diagnostic process, not necessarily in the outcome. Applied to legislative design: approving a high-impact norm without a Regulatory Hysteresis Impact Statement estimating its RHI constitutes the same omission. The state cannot argue that the norm was valid, well-intentioned, or even beneficial in net terms if it failed to evaluate the systemic risk before creating it.

Omission 2: Not Using Ecography or X-ray When Available (Ignoring Simulation Tools)

A physician who diagnoses without imaging when symptoms indicate the need and imaging is available is negligent. The tool's availability is what creates the obligation. Applied to legislative design: from February 2026, evolutionary simulation tools are documented as available and reliable for assessing institutional dynamics. Failing to apply them to high-RHI proposed norms is the regulatory analog of refusing to use imaging when the clinical indication is clear.

Omission 3: Administering Anesthesia Without Consulting Patient History (Approving Norms Without Evaluating Existing Institutional Phenotype)

A physician who administers anesthesia without consulting the patient's history, including prior reactions, current medications, and contraindications, is negligent regardless of the anesthesia's intrinsic quality. Applied to legislative design: approving a norm in a domain with high cumulative RHI (high institutional deformation already present) without evaluating how the new norm will interact with existing extended phenotypes is the regulatory analog of anesthesia without history. The existing institutional infrastructure is the patient's medical history. Its omission from the legislative analysis constitutes independent negligence.

6.5 The Temporal Threshold: A Precise Date for the New Standard

Legal standards of care typically evolve gradually, making it difficult to identify the precise moment when a new tool becomes part of the lex artis. The publication of Shapira et al. (2026) provides an unusually precise inflection point for three reasons.

First, the paper establishes in peer-reviewed form, from multiple elite research institutions, that synthetic agent simulations reliably replicate institutional dynamics. This is not a preliminary finding or a theoretical proposal; it is empirical documentation across eleven case studies with reproducible methodology.

Second, the paper explicitly invites interdisciplinary engagement from legal scholars and policymakers. The abstract states that the findings 'warrant urgent attention from legal scholars, policymakers, and researchers across disciplines.' The community of legislative designers has been directly addressed.

Third, the underlying infrastructure (OpenClaw, LLM-powered agents, evaluation protocols) is open-source and accessible, not proprietary technology limited to well-resourced actors. The elevator is not only available; it is free to use.

From February 23, 2026 onward, the omission of evolutionary simulation analysis for proposed norms with RHI > 0.50 falls below the standard of diligence that a reasonably informed legislative process should satisfy. Norms approved after that date without documented simulation analysis carry a rebuttable presumption of negligent process in hysteresis damage litigation.

VII. State Liability: From Binary to Graduated Strict Liability

7.1 The Inadequacy of the Current Framework

State liability for legislative activity in Argentina operates under Law 26,944 (2014), which requires for legitimate state activity a 'special sacrifice,' 'certain and actual damage,' direct and exclusive causal chain, and explicitly excludes lost profits. For illegitimate activity, it requires a 'service failure.' Neither category captures what regulatory hysteresis produces.

The damage from hysteresis is not special: it affects the entire regulated population, not a specific individual. It is not always certain and actual at enactment: it materializes progressively as the extended phenotype solidifies. The activity is not illegitimate in the formal sense: the norm was validly enacted. And the binary logic of legitimate/illegitimate fails to capture the negligent process defect that is the actual source of liability.

Public law's binary thinking about legislative liability is institutionally adaptive (it protects legislative autonomy and democratic legitimacy) but theoretically inadequate when the damage comes not from the norm's content but from negligence in its production process. A norm can be perfectly constitutional, formally valid, even beneficial in expected value, and still generate irreversible institutional damage because it was produced without quality controls.

7.2 Strict Liability for Risk Creation: Arts. 1757-1758 CCyCN

The Civil and Commercial Code (2015) provides the appropriate conceptual foundation. Articles 1757 and 1758 establish strict liability for risk creation: whoever develops a risky activity or uses risky things is liable for resulting damages without need to prove fault or mental states. The created risk is sufficient.

Legislating is a risky activity in precisely the sense these articles contemplate. The risk is not merely that the norm will be applied incorrectly; it is that the legislative process, conducted without quality controls, will generate extended phenotypic infrastructure that is irreversible, imposes forced shared use on all regulated subjects, and accumulates in ways that destroy future option value. This risk is inherent to the activity of normative production and does not depend on demonstrating the specific subjective fault of any legislative actor.

The state, as the institutional guardian of legislative activity, should bear strict liability for the institutional damage generated, subject to two defenses: first, that it incorporated adequate quality controls into the production process (hysteresis impact assessment, evolutionary simulation for high-RHI norms, sunset clauses with PDCA cycles, outcome metrics); second, that the urgency of the circumstances precluded the delay required for standard process (an emergency exception analogous to emergency surgery without full pre-operative workup).

7.3 Graduated Liability According to Intentionality Level

The strict liability framework does not produce undifferentiated joint liability across all participants in the legislative chain. Liability is graduated according to the intentionality level demonstrated by each actor and the tools available to them at the relevant time.

Individual legislators (Level 3 agents) bear proportional responsibility for foreseeable hysteretic damage when they voted with actual or constructive knowledge that the norm would generate irreversible institutional deformation. After February 2026, constructive knowledge is established for any norm with RHI > 0.50 in a domain where simulation tools were available and not consulted.

Committee chairs and legislative leaders bear greater responsibility than backbench members because they directed the consideration process and had the institutional authority to require hysteresis impact assessment. Their responsibility is both for what they did and for what they failed to require.

The executive branch, as designer and sponsor of legislative initiatives, bears design responsibility for the normative product proportional to the predictability of the hysteretic damage at the time of submission. For initiatives submitted after February 2026 with RHI > 0.50, the failure to conduct and document evolutionary simulation analysis constitutes negligent design.

Technical legislative advisors bear professional liability for failing to inform decision-makers of available quality tools, particularly when those tools are directly applicable to the norm under consideration and have been documented in the scientific literature accessible to professionals in the field.

7.4 The Francovich Extension: Domestic Hysteretic Damage

The Francovich/Brasserie doctrine of the European Court of Justice recognizes state liability for legislative acts violating Community law when three conditions are met: a norm conferring rights, a sufficiently grave violation, and a direct causal nexus. This doctrine has no direct equivalent in Argentine domestic law, but its logical structure can be adapted for hysteretic damage claims.

Francovich Requirement	Adaptation for Domestic Hysteretic Damage
Norm conferring rights	Constitutional right to economic freedom and regulatory reasonableness (Arts. 14, 28 CN); right to legislative process with adequate quality controls
Sufficiently grave violation	RHI > 0.70 without documented hysteresis assessment; omission of evolutionary simulation for high-impact norm after Feb. 23, 2026
Direct causal nexus	Demonstrated compliance cost exceeding marginal regulatory benefit; measurable destruction of option value; quantifiable residual deformation attributable to the specific norm

The proof of damage shifts under this framework. There is no need to demonstrate 'certain and actual' damage at the time of enactment. Hysteresis is foreseeable and quantifiable ex ante using the RHI. The omission of that evaluation is the negligence. The damage is the difference between the hysteresis produced and the hysteresis that would have been produced with adequate quality controls and, where applicable, evolutionary simulation analysis.

VIII. Seven Falsifiable Predictions

A theoretical program earns scientific status through the specificity of its falsifiable predictions. The integrated framework developed in this paper generates seven.

Prediction 1 (Phenotype-Hysteresis Proportionality): Hysteresis magnitude is proportional to the density of the extended phenotype generated. Norms that create specialized jurisdiction, licensed professionals, and academic doctrine generate more residual deformation than merely procedural norms of comparable regulatory scope. Measurement: compare the institutional cost of repealing norms with dense extended phenotype versus norms with sparse extended phenotype, controlling for regulatory scope and political salience.

Prediction 2 (RIA Conditionality on Reversibility Analysis): Jurisdictions with formal regulatory impact assessment mechanisms that include explicit reversibility analysis should exhibit lower cumulative hysteresis than jurisdictions with standard RIA or none. Standard cost-benefit RIA that does not address reversibility would not capture the effect. Measurement: cross-national comparison of cumulative RHI in jurisdictions with reversibility-inclusive RIA versus those without, controlling for regulatory volume and institutional development.

Prediction 3 (Annealing Threshold): There exists a threshold of cumulative hysteresis beyond which incremental reform becomes impossible without exogenous institutional annealing. Prediction: this threshold falls between RHI 0.80 and 0.90 across democratic systems. Measurement: identify the RHI level above which no sustained reform has succeeded across a sample of 10+ countries, using the historical reforms database developed in Lerer 2025b.

Prediction 4 (Extractive Coalition Correlation): In jurisdictions with RHI > 0.70, the proportion of GDP devoted to compliance should correlate positively with resistance to deregulatory reform, measured by time from introduction to committee vote for simplification bills. The causal mechanism is the Olson dynamic: compliance beneficiaries mobilize against the bills that would reduce their institutional rents. Measurement: cross-national or cross-sectoral analysis using compliance cost data (OECD 2025) and legislative processing time for regulatory simplification initiatives.

Prediction 5 (Option Value Destruction): Jurisdictions that adopt sunset clauses for high-RHI norms should demonstrate greater regulatory diversity (more distinct approaches to similar policy problems) than jurisdictions without sunset clauses, controlling for other variables. The mechanism is that compulsory periodic review re-opens the option space that hysteresis had closed. Measurement: comparative analysis of regulatory approach diversity using OECD regulatory inventory data.

Prediction 6 (Simulation-RHI Reduction): Norms approved after February 2026 with documented evolutionary simulation analysis should exhibit RHI values significantly lower than norms of comparable institutional impact approved without such analysis, controlling for regulatory domain, political context, and legislative coalition composition. The mechanism is that pre-legislative simulation identifies high-hysteresis design features that can be modified before enactment. This prediction is prospective and will require a minimum 5-year observation window to test.

Prediction 7 (Parasitic Meme Drift): Norms that persist for more than 20 years without mandatory review should exhibit increasing divergence between their declared protective rationale and their measurable beneficiary distribution. The mechanism is the parasitic meme dynamic: as the originating norm's extended phenotype solidifies, it develops selective pressure for interpretations that favor the institutional infrastructure's maintenance over the norm's stated welfare objective. Measurement: comparative analysis of stated and effective beneficiary distributions for long-standing norms across multiple domains, using benefit incidence analysis.

IX. Policy Implications

9.1 Pre-Legislative Hysteresis Impact Statement

Every bill with projected RHI > 0.30 should include a mandatory Regulatory Hysteresis Impact Statement estimating the five dimensions of the base RHI and the destroyed option value component. This statement should be produced by technical advisors independent from the sponsoring legislative bloc and should address explicitly: what jurisdictional structures the norm creates; what professional specialization it induces; what doctrinal elaboration is expected; what international obligations it generates or reinforces; what administrative infrastructure it requires; and what regulatory alternative spaces it forecloses.

For norms with projected RHI > 0.50, the Impact Statement should include documented results of evolutionary simulation analysis using available synthetic agent tools. The simulation should test at minimum: stability of the norm's cooperative equilibrium under adversarial conditions; whether the norm's design is an ESS or is susceptible to invasion by evasion strategies; whether lock-in in suboptimal equilibria emerges; and what second-order institutional effects develop over extended simulation periods.

9.2 Sunset Clauses with PDCA Cycles

High-RHI norms should include mandatory review clauses tied to measurable outcome indicators, implementing the Plan-Do-Check-Act cycle that Deming articulated for manufacturing. Empirical evidence on regulatory sunset provisions (analyzing effectiveness across U.S. states) establishes that sunset provisions are the most effective mechanism for reducing regulatory levels, surpassing RIA requirements and other review mechanisms. The mechanism is institutional annealing: periodic mandatory reviews introduce perturbations that prevent the system from converging irreversibly to suboptimal attractors.

Current sunset clauses fail because they lack the Check and Act phases: they mandate re-examination but do not require outcome measurement against original specifications nor adjustment based on evidence. Effective sunset architecture for high-RHI norms must include: explicit outcome metrics defined at enactment; mandatory measurement against those metrics at review; binding obligation to act on measurement results (renew, amend, or repeal); and assessment of accumulated RHI growth since enactment.

9.3 Graduated State Liability as Incentive Mechanism

Dari-Mattiacci, Garoupa, and Gomez-Pomar identify three distinct functions of state liability: providing incentives to the state to take precautions, removing perverse incentives to private actors who might otherwise under-invest in self-protection, and enabling citizen monitoring of state activity. A graduated strict liability framework based on RHI and intentionality levels would activate all three functions simultaneously.

The incentive effect operates at the design stage: knowing that high-RHI norms without simulation analysis carry a rebuttable presumption of negligent process will induce executive branches to incorporate evolutionary assessment in their normative production. The monitoring function operates through the RHI as a publicly computable index: civil society, academic institutions, and affected parties can calculate the hysteresis risk of proposed norms before enactment and invoke the liability framework if adequate process was not followed. The private incentive function operates by establishing that regulated actors who demonstrate harm from hysteretic deformation can recover, reducing the moral hazard of assuming the state will eventually compensate without requiring preventive action.

9.4 Institutional Annealing for Maximum-Hysteresis Regimes

Where cumulative RHI exceeds the annealing threshold of approximately 0.85 (Prediction 3), incremental reform has demonstrated zero success rate. At this level, the institutional equivalent of metallurgical annealing is required: an exogenous perturbation strong enough to disrupt the existing attractor and allow the system to reconverge to a new equilibrium. Historical examples include the German and Japanese constitutional transitions of 1947-1949 and, partially, Chile's 1980 constitutional framework.

Democratic annealing differs from imposed constitutional transition. It requires simultaneous intervention across multiple dimensions of the extended phenotype: jurisdictional restructuring, professional transition programs that reduce the economic cost of despecialization, revision of international treaty obligations, administrative procedure reform, and coordinated academic curriculum change. The RHI provides the measurement tool to assess whether annealing has been achieved or whether residual deformation persists.

X. Conclusion

The three theoretical traditions integrated in this paper converge on a single insight***: regulatory production is a manufacturing process that operates without quality controls, generates permanent institutional deformation through the extended phenotype mechanism, and creates damage for which no adequate liability framework currently exists***.

The absence of quality criteria in normative production is not an accidental omission. It is a structural consequence of memetic dynamics: legal memes do not optimize the functional quality of the regulatory product; they optimize their own replication. A norm of poor functional quality can be perfectly successful in memetic terms if it generates jurisprudence, doctrine, specialized professorships, and professional litigation. The functional quality of regulation and the memetic fitness of the underlying legal concept are independent variables, sometimes inversely correlated.

Regulatory hysteresis provides the missing concept that connects institutional description (the extended phenotype) **to institutional prescription** (how normative production should be managed to minimize irreversible damage, and what happens when it is not). The Regulatory Hysteresis Index, the multilevel intentionality liability framework, and the seven falsifiable predictions constitute a research program that can be tested against cross-national data and, if validated, can inform concrete institutional reform.

The publication of Shapira et al. (2026) adds a time-stamped inflection point to that program. Until February 23, 2026, the argument that no tools existed to test institutional dynamics before implementing high-impact norms retained some force as a liability defense. After that date, the argument is unavailable: the scientific community has demonstrated, across eleven case studies in a controlled laboratory environment, that synthetic agent populations with evolutionary time replicate the dynamics that EPT predicts for human institutional systems. The lex artis applicable to normative production has changed.

What remains is whether legal systems, including the Argentine one with its CLI of 0.87 and its 0/23 reform record, will recognize the change. The elevator is available. The stairs were acceptable when the elevator was under construction. They are not acceptable now.

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Research Methodology Note

This paper was written entirely by the author. For mathematical formalization of certain passages, consistency verification across cross-referenced frameworks, and literature review assistance, generative AI systems (Claude, Anthropic; Genspark; Perplexity) were used, in accordance with emerging practices of transparency in AI-assisted academic research. The theoretical frameworks (EPT, RHI, multilevel intentionality liability, the lex artis argument regarding Shapira et al. 2026), all empirical observations, and all legal interpretations are independent products of the author's research program. All errors are the author's.